

# CS & IT ENGINEERING



**THEORY OF COMPUTATION**

## Grammar

Lecture - 03



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# Topics to be covered



Topic

Grammar Construction  
?????

Types of Grammar

Ambiguous Grammar  
CFG, Simplification, Normal form





# { Ambiguous Grammar }

A Grammar is ambiguous for at least one string if there exist more than one

✓ Leftmost Derivation

✓ Rightmost Derivation

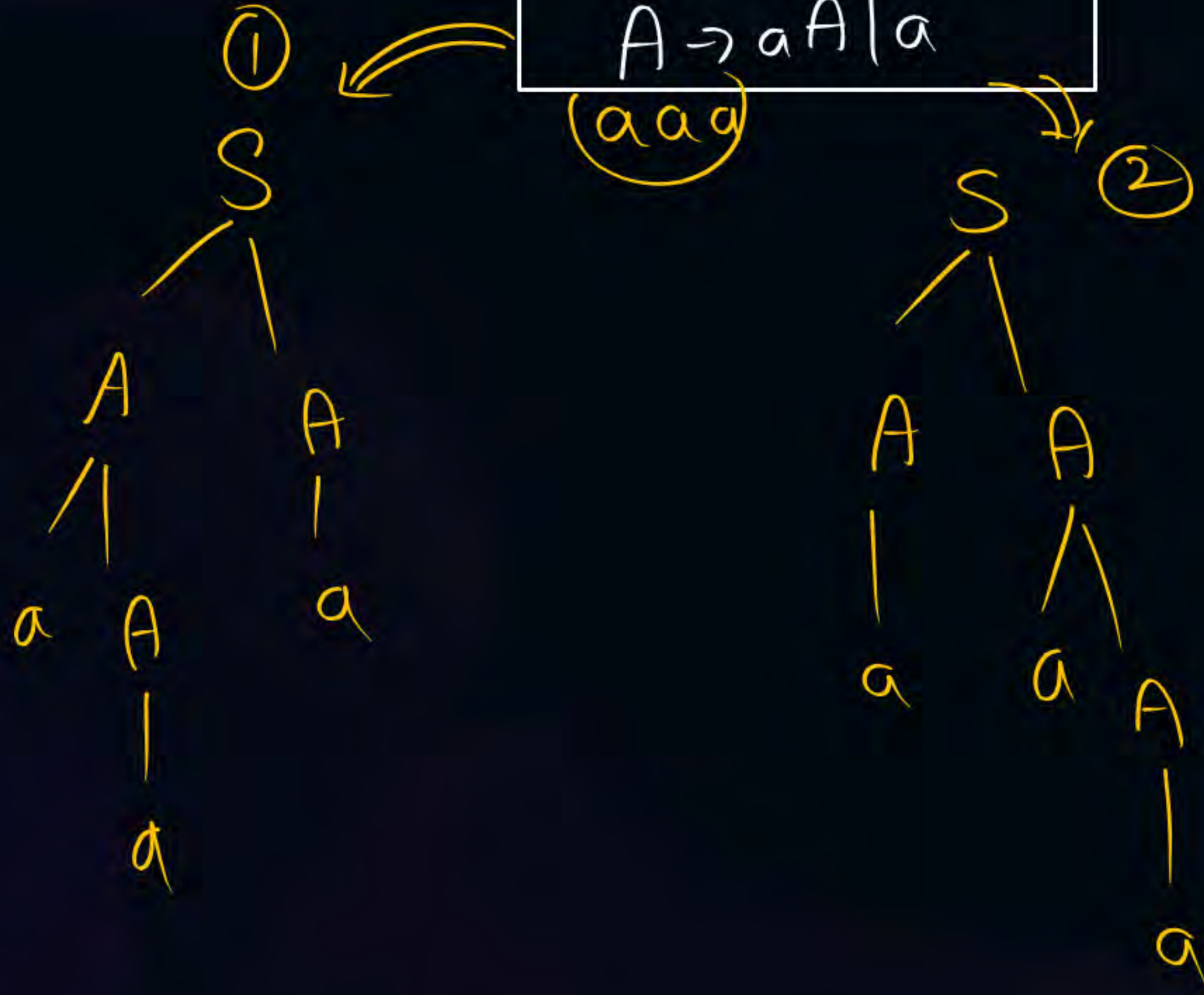
✓ Parse tree ✓



Q) Verify given Grammar is ambiguous or not?

$$\begin{aligned} S &\rightarrow AA \\ A &\rightarrow aA \mid a \end{aligned}$$

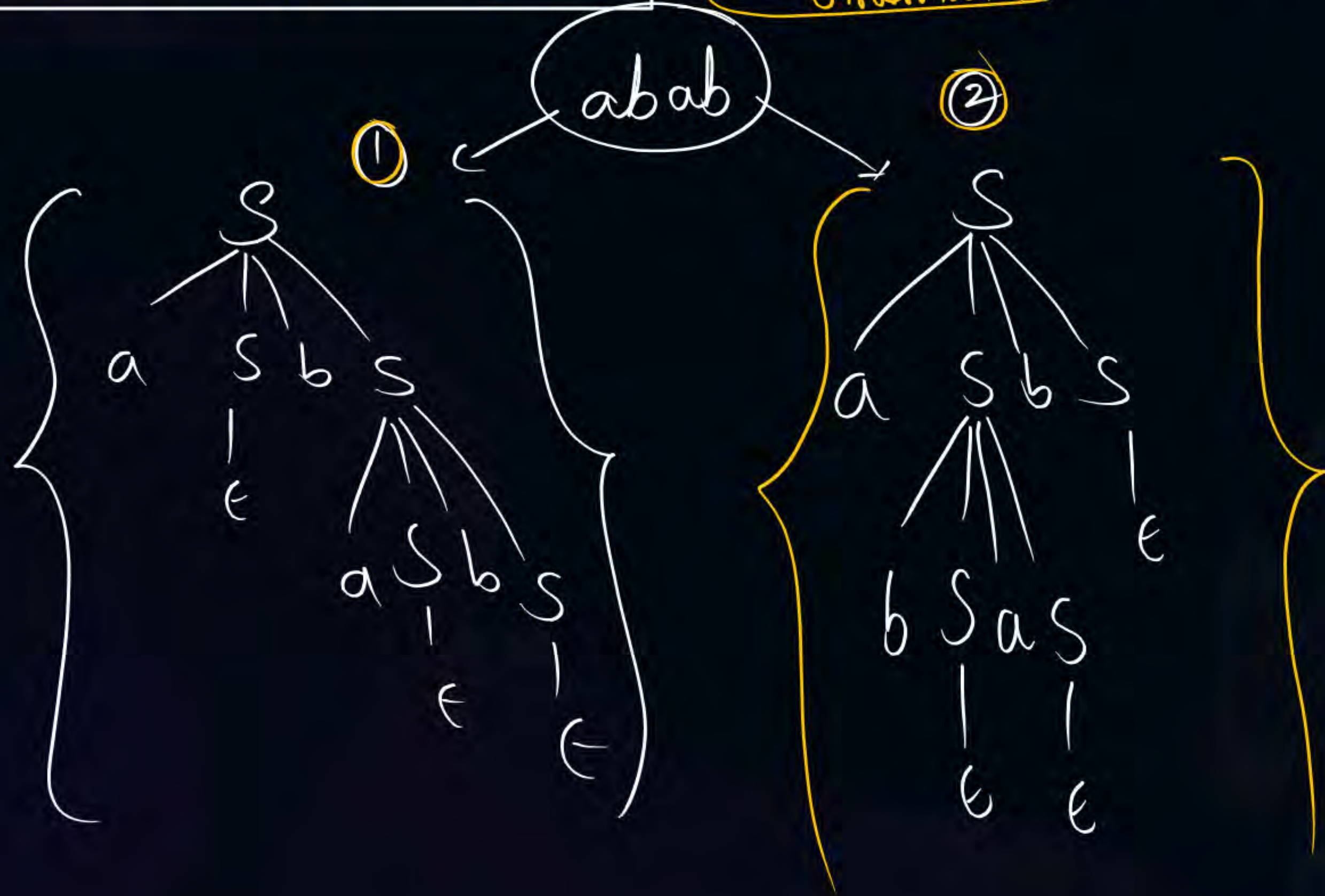
Ambiguous



②

$S \rightarrow aSbS / bSaS / \epsilon$

Ambiguous or not?





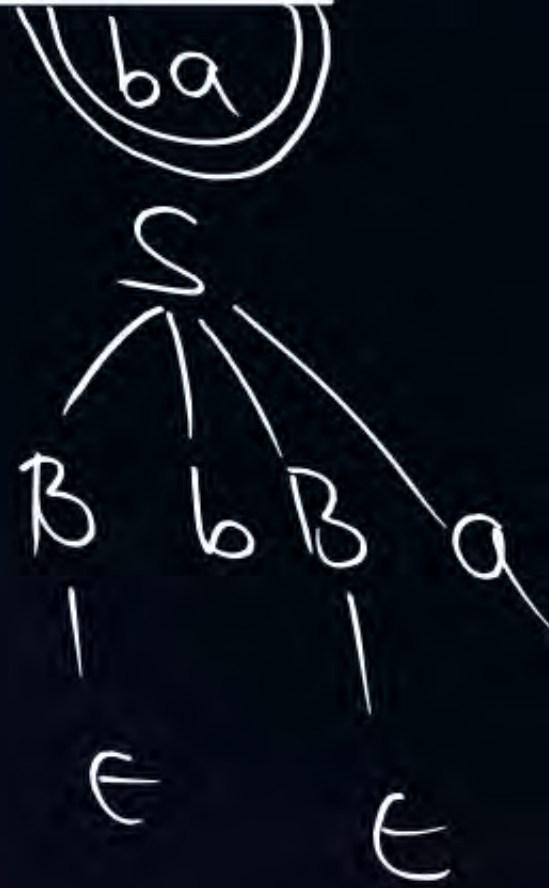
③

$$S \rightarrow AaBb \mid BbBa$$

$$A \rightarrow \epsilon$$

$$B \rightarrow \epsilon$$

Unambiguous Grammar



$$\textcircled{2} \quad E \rightarrow E + T \mid T$$

$$T \rightarrow T * F \mid F$$

$$F \rightarrow a$$



## NOTE



① Checking Ambiguity is Undecidable Problem

[No Algorithm exist to check ambiguity in Grammar]

② Elimination of ambiguity from grammar is also Undecidable Problem.



# Ambiguous Grammar

$$E \rightarrow E + E \mid E * E \mid n$$

input string

$$n + n * n$$

$$2 + 3 * 4$$

$$2 + 12$$

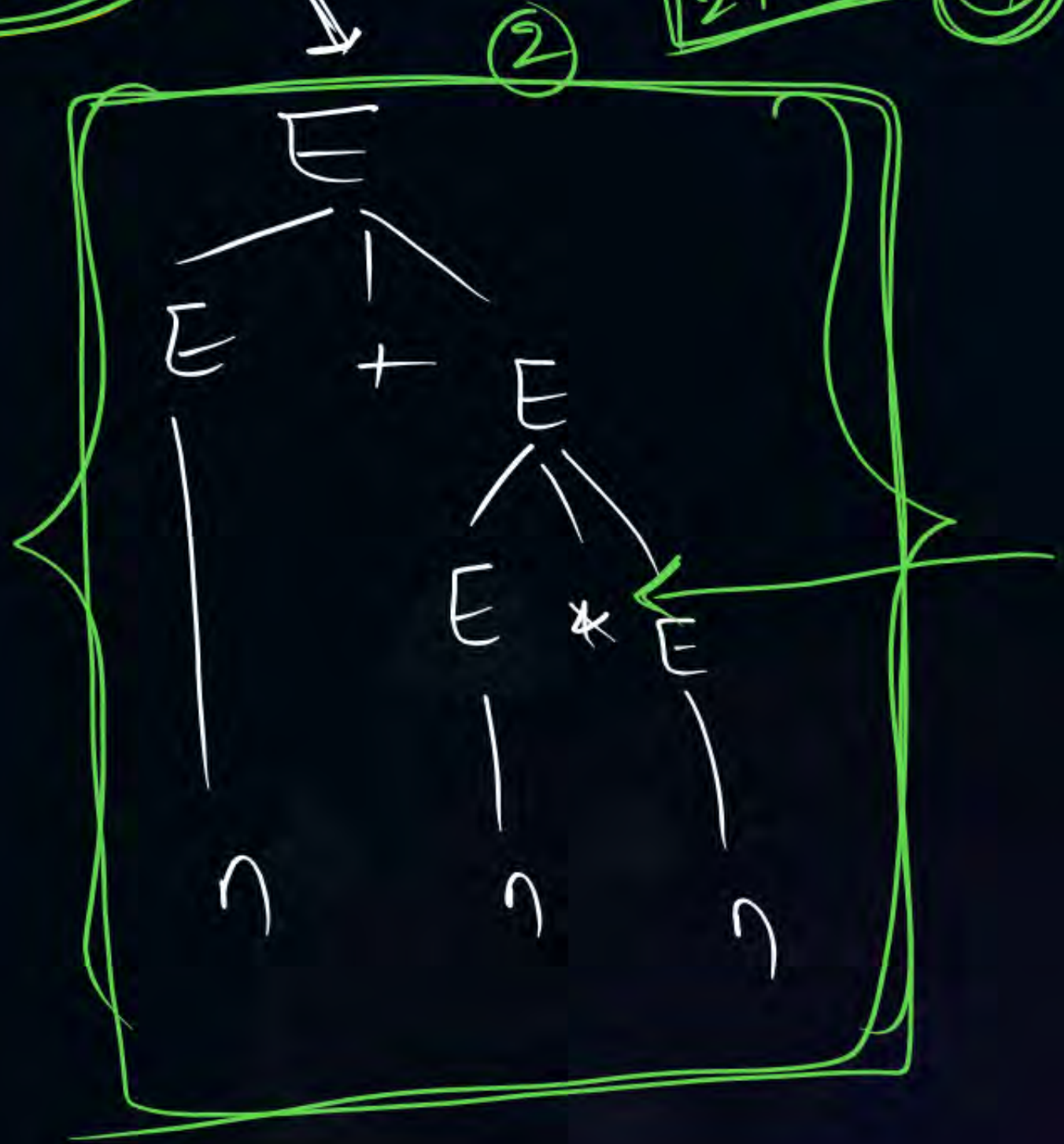
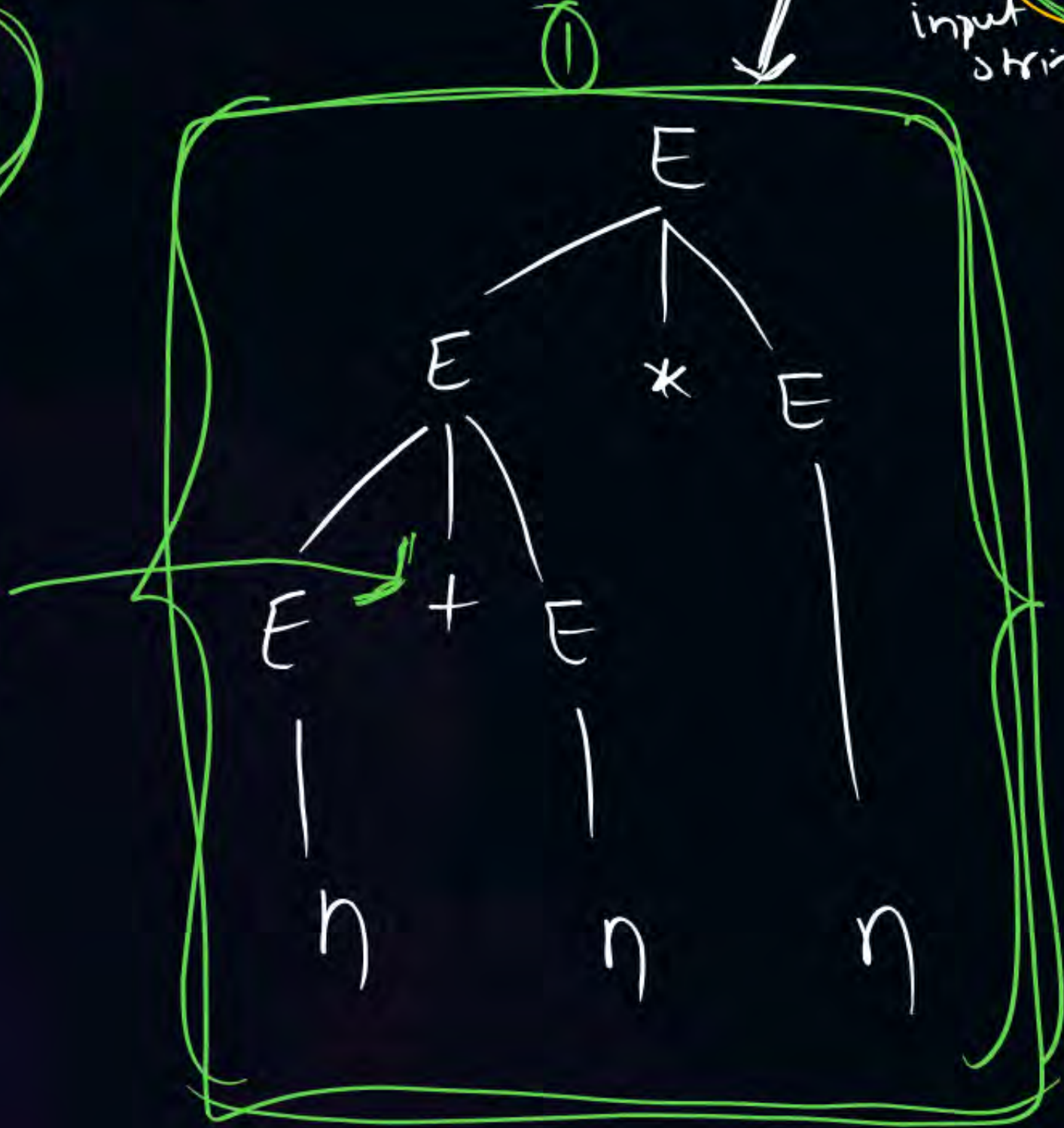
$$(14)$$



$$2 + 3 * 4$$

$$5 * 4$$

$$(20)$$





# Simplification of CFG



① Elimination of useless Variables

② Elimination of Unit Productions ( $A \rightarrow B$ )

③ Elimination of null Productions  
( $A \rightarrow \epsilon$ )



①

$$S \rightarrow aS, b$$

$$S_1 \rightarrow aS, b | \epsilon$$

$$S \rightarrow aS, b | ab$$

$$S_1 \rightarrow aS, b | ab$$



②

$$S \rightarrow AaB$$

$$A \rightarrow a | \epsilon$$

$$B \rightarrow b | \epsilon$$

Remove Null Productions

$$S \rightarrow AaB | aB | Aa | a$$

$$A \rightarrow a$$

$$B \rightarrow b$$



Ex

1)

$S \rightarrow A$

$S \rightarrow a$

2)

$A \rightarrow B$

$A \rightarrow a$

3)

$B \rightarrow C$

$B \rightarrow a$

4)

$C \rightarrow D$

$C \rightarrow a$

$D \rightarrow a$

$D \rightarrow a$

Unit Productions

useless

$\{A, B, C, D\}$



## Elimination of Useless Variable

- 
- ① Eliminate Variables which are not producing any string
  - ② Eliminate variables which are not required for derivation



(Q) How many Productions Present in Simplified Grammar

$$\begin{array}{l}
 S \rightarrow \boxed{AB} | a \\
 A \rightarrow a \\
 \text{p } B \rightarrow aB | bC \\
 \text{p } C \rightarrow aB | aC \\
 E \rightarrow a
 \end{array}$$

$$S \rightarrow a \quad \left\{ \begin{array}{c} S \\ \downarrow \\ a \end{array} \right\}$$

$$\boxed{A \rightarrow a} \\
 \boxed{E \rightarrow a} \quad \times$$

$$\boxed{S \rightarrow a}$$

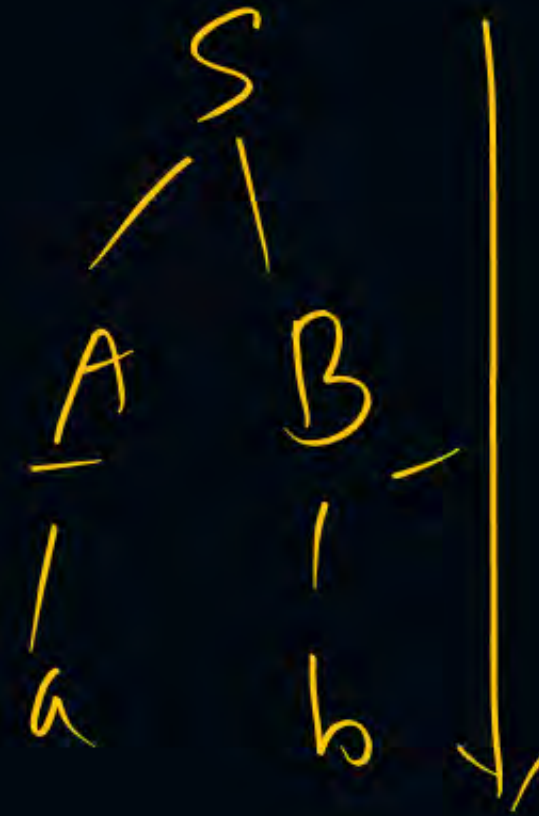
①



$$S \rightarrow AB$$

$$A \rightarrow a$$

$$B \rightarrow b$$

$$E \rightarrow a$$




①

$$\begin{aligned} \checkmark S &\rightarrow AB \mid b \\ \checkmark A &\rightarrow a \\ \times B &\rightarrow bB \end{aligned}$$

B is useless variable

②

$$\begin{aligned} \times S &\rightarrow \overbrace{AB}^{\emptyset} \\ A &\rightarrow a \\ \times \boxed{B \rightarrow bB} &= \emptyset \end{aligned}$$

(S, B)

**THANK - YOU**